



Introduction to Photogrammetry

Lecture 11

Introduction and Classification of Aerial Photographs

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In Today's Class

1. Introduction
2. Classification of Aerial Photographs

1. Introduction

- Aerial photography refers to photographs of the Earth's surface taken from an airborne platform such as an aircraft, drone, or satellite.
- It is one of the most fundamental data sources in photogrammetry and remote sensing because it allows us to observe and measure the Earth from above.

- Aerial photographs are widely used for:
 - Topographic mapping
 - Urban planning and monitoring
 - Environmental studies
 - Disaster management
 - Military reconnaissance
- They provide both **qualitative interpretation** and **quantitative measurement** capabilities.

Factors Affecting Quality of Aerial Photographs

- The quality of aerial photographs depends on several factors, including:
 - The design and calibration of the camera system
 - The quality of lenses and sensors
 - The type of photographic film or digital sensor
 - Atmospheric conditions such as haze and clouds
 - Stability of the aircraft during image capture

2. Classification of Aerial Photographs

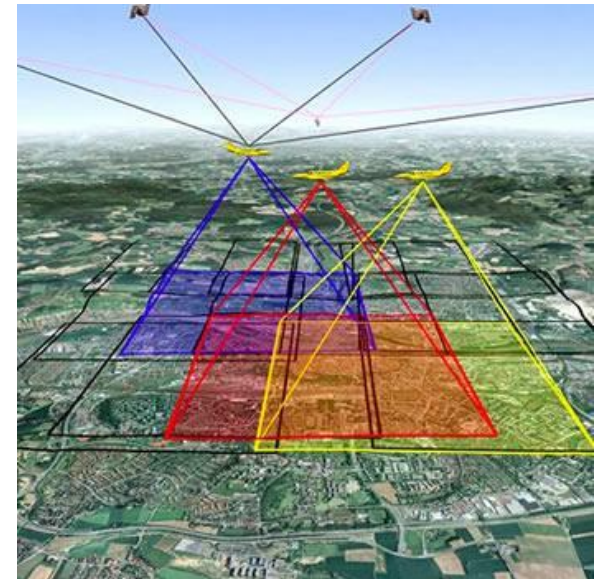
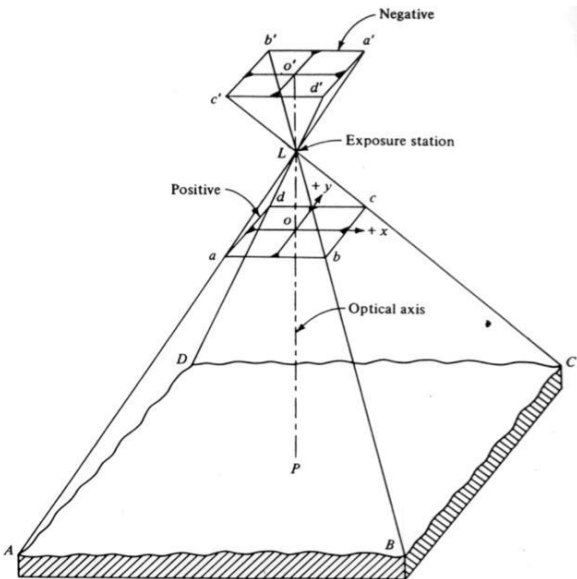
- Aerial photographs can be classified based on:
 - A. Orientation of the camera axis
 - B. Angular coverage (field of view)
 - C. Type of photographic emulsion
- Each classification affects how the image is interpreted and used.

A. Orientation of Camera Axis

- The orientation of an aerial photograph depends on the angle between the **camera axis** and the vertical direction.
- This classification is extremely important for mapping accuracy.
- This classification is divided into two types:
 - i. Vertical photographs
 - ii. Oblique photographs

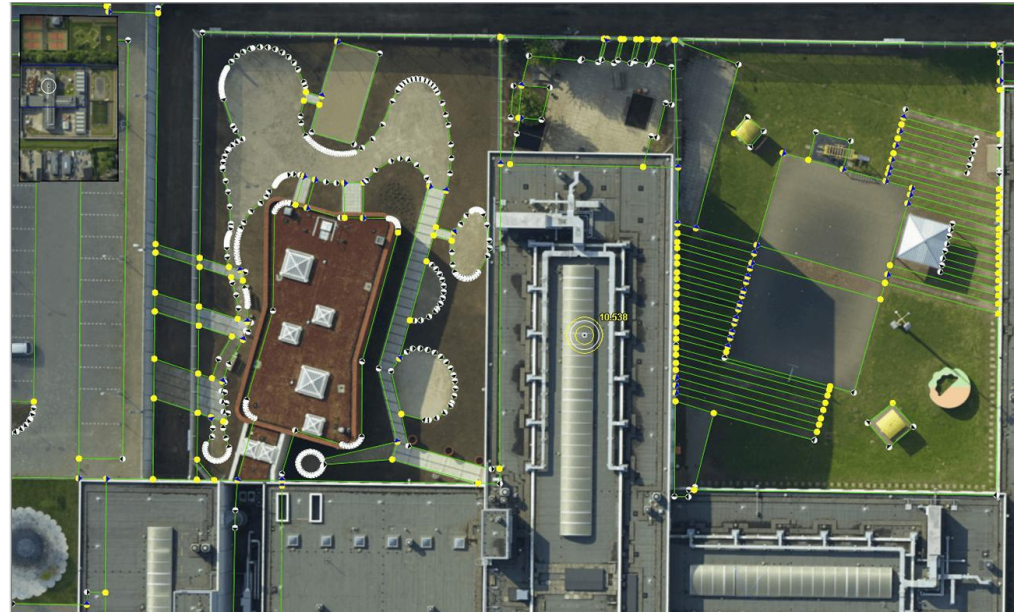
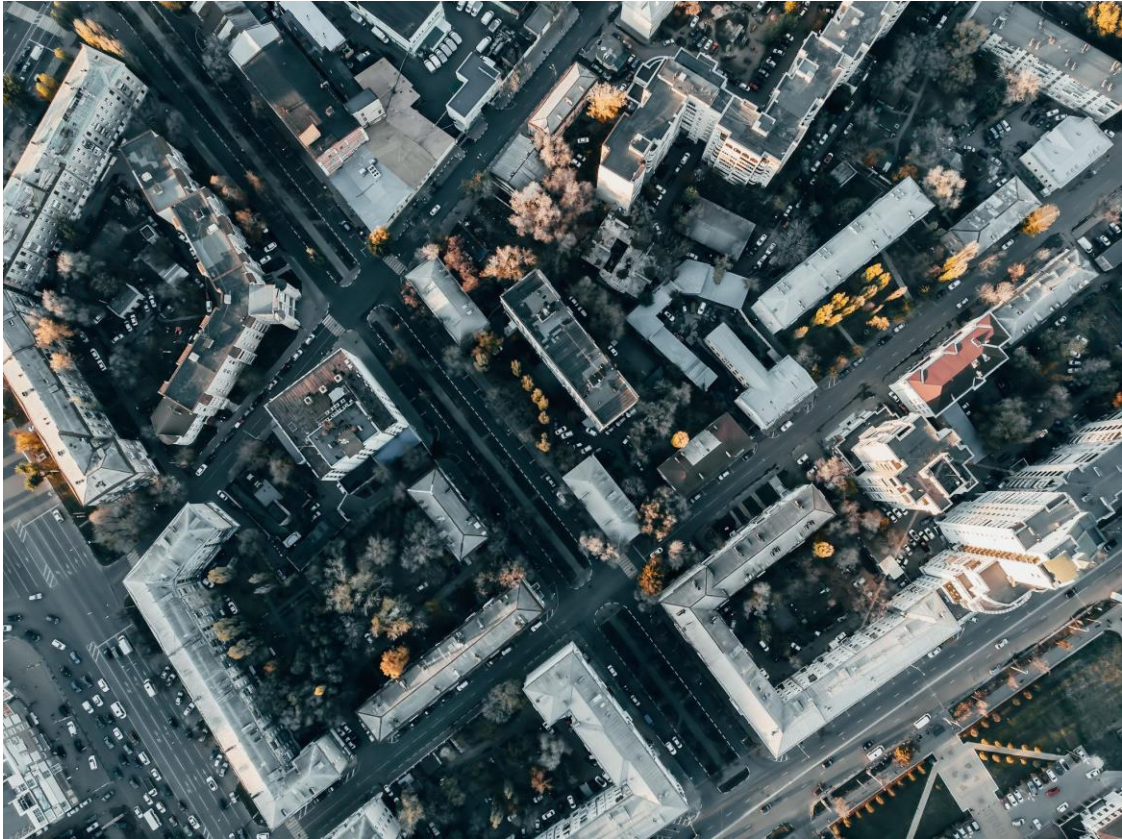
i. Vertical Photograph

- A vertical photograph is taken when the camera axis is nearly perpendicular to the ground surface.
- In practice, a small tilt (usually less than 3°) is acceptable.



Characteristics of Vertical Photographs

- Vertical photographs have the following characteristics:
- They have nearly uniform scale across the image
- Distortion is minimal
- They are most suitable for mapping and measurements



ii. Oblique Photograph

- An oblique photograph is taken when the camera axis is intentionally tilted away from the vertical.
- These photographs provide a more natural perspective view.
- Oblique photographs are divided into two types:
 - a) Low oblique photographs
 - b) High oblique photographs

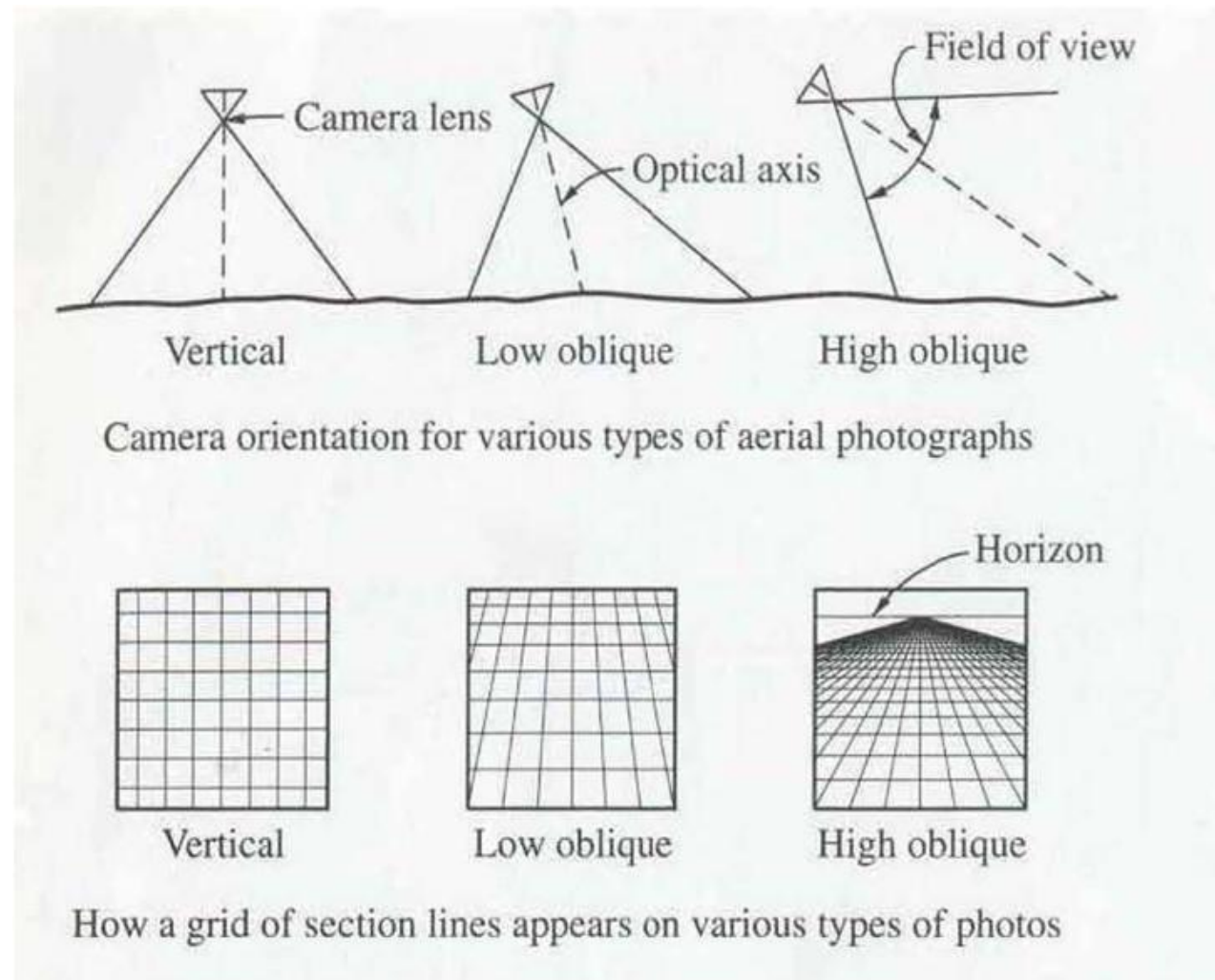
Low Oblique Photograph

- In a low oblique photograph:
- The camera is tilted slightly
- The horizon is not visible
- These are useful for local area interpretation.

High Oblique Photograph

- In a high oblique photograph:
- The camera is tilted at a larger angle
- The horizon is clearly visible
- These images provide a broader perspective.





Comparison: Vertical vs Oblique

- Vertical photographs are ideal for **measurement and mapping**, whereas oblique photographs are more suitable for **visual interpretation and reconnaissance**.

B. Angular Coverage

- Angular coverage refers to the extent of the ground area captured by the camera lens.
- It is determined by the **field of view (FOV)** of the camera.
- Aerial photographs are classified into:
 - i. Narrow-angle photographs
 - ii. Normal-angle photographs
 - iii. Wide and Super wide-angle photographs

i. Narrow Angle

- Narrow-angle photographs capture a smaller area with less distortion.

ii. Normal Angle

- normal-angle photographs provide a balance between coverage and distortion.

iii. Wide and Super Wide Angle

- Wide-angle and super wide-angle photographs cover larger areas, but they introduce more distortion, especially near the edges.

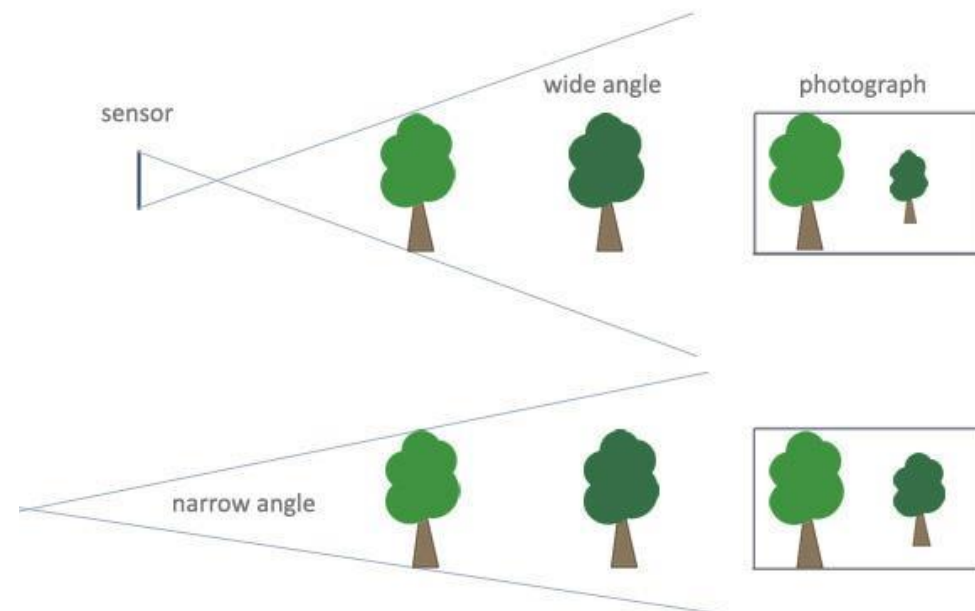


Image quality in the middle



Image quality on the edge

C. Emulsion Type

- The emulsion type refers to the type of film or sensor used to capture the image, which determines how different wavelengths of light are recorded.
- This classification includes;
 - i. Panchromatic Photography
 - ii. Color Photography
 - iii. Infrared Photography

i. Panchromatic Photography

- Panchromatic images record the visible spectrum and appear in black and white. They are widely used due to their high spatial resolution.

ii. Color Photography

- Color photographs represent objects in their natural colors, making them easier to interpret visually.

iii. Infrared Photography

- Infrared photography records reflected infrared radiation. It is particularly useful for vegetation and environmental analysis.





Thank you!

Any Questions
